

Exponential ADSR Envelope Generators & Peak Trackers

Mathematical Formalism of [adsr~] and [env~] DSP Nodes

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Abstract

This document formulates the piecewise exponential state machine governing 4-stage ADSR envelope generators ([adsr~]) and the continuous peak/RMS energy tracking equations of audio envelope followers ([env~]).

1 Analog Exponential RC Charging Curves

Analog envelope generators utilize capacitor charging across resistors, resulting in natural exponential curves. The continuous-time voltage response $v(t)$ when approaching target voltage V_{target} from initial voltage V_0 is:

$$v(t) = V_{\text{target}} + (V_0 - V_{\text{target}}) e^{-t/\tau_c} \quad (1)$$

where $\tau_c = R \cdot C$ is the RC time constant.

2 Discrete-Time ADSR State Machine

In sample domain at rate f_s , the per-sample update for envelope level $e[n]$ is:

$$e[n] = e[n-1] + \alpha_{\text{stage}} \cdot (V_{\text{target, stage}} - e[n-1]) \quad (2)$$

where the smoothing coefficient $\alpha_{\text{stage}} \in (0, 1)$ is determined by the desired time duration T_{stage} (in seconds):

$$\alpha_{\text{stage}} = 1 - \exp\left(-\frac{1}{f_s \cdot T_{\text{stage}}}\right) \quad (3)$$

2.1 Stage Definitions

The finite state machine switches states based on threshold completion:

1. **Attack Stage** ($e[n-1] < 1.0$): $V_{\text{target}} = 1.2$ (overshooting to ensure linear-like fast rise), switching to Decay when $e[n] \geq 1.0$.
2. **Decay Stage** ($e[n-1] > S$): $V_{\text{target}} = S$ (Sustain level $S \in [0, 1]$), transitioning to Sustain when $|e[n] - S| < \epsilon$.
3. **Sustain Stage**: $e[n] = S$ held constant while gate is active (gate > 0).
4. **Release Stage** (gate $== 0$): $V_{\text{target}} = -0.05$ (undershooting target to reach absolute zero quickly), terminating when $e[n] \leq 0.0$.

3 Audio Envelope Follower ([env~]) Peak/RMS Tracker

The [env~] node extracts signal amplitude envelope $A[n]$ from input audio $x[n]$ using asymmetrical attack/release smoothing coefficients:

$$|x[n]| = \text{rectify}(x[n]) = \sqrt{x[n]^2 + \delta} \quad (4)$$

$$A[n] = \begin{cases} (1 - \alpha_{\text{attack}})A[n-1] + \alpha_{\text{attack}}|x[n]|, & |x[n]| > A[n-1] \\ (1 - \alpha_{\text{release}})A[n-1] + \alpha_{\text{release}}|x[n]|, & |x[n]| \leq A[n-1] \end{cases} \quad (5)$$

Converting amplitude $A[n]$ to decibels RMS (dB_{SPL}):

$$L_{\text{dB}}[n] = 20 \log_{10} (\max(10^{-4}, A[n])) \quad (6)$$